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The Importance of Randomness to Quantum Error Correction Algorithms and Quantum Error Correction Algorithms Applicable to IBM Quantum Computer

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Abstract

Until now there is no perfect quantum computer system, and the qubits are always with errors. As a result, error correction becomes extremely challenging, requiring fundamentally different approaches than traditional methods.In our study, we use a simulator to implement Shor's error correction code (SRCC) and develop Shor's error correction code with random method (SRCC-R) on bit flip errors problem. We consider the IBM quantum computer system with the error rate 0.007 and 0.002. In our result shows that using SRCC the flip errors can correct less than 25 controlled-not gates with 0.97 correct rate. In our method, SRCC-R the flip errors can correct larger than 500 controlled-not gates with 0.95 correct rate.

SHOR'S ERROR CORRECTION CODE

Shor's Error Correction Code, proposed by Peter Shor in 1995, is a quantum error correction method. It distributes the information of one qubit across multiple qubits and uses redundancy to detect and correct errors during quantum computation. This algorithm inspired the concept discussed in this article. Additionally, in 1996, Steane discovered a 7-qubit general code, and the general theory of quantum error correction was presented by Calderbank and Shor, and Steane.

REASONS FOR ERROR CORRECTION

This study focuses on Bit Flip errors for several reasons:

1. Bit information is crucial in quantum computation.
2. While phase information is important, the same error-correction method can be applied to phase information by changing the measurement basis.
3. We aim to assess the quantum computer's ability to handle errors, without considering hardware-level methods for reducing errors or enhancing fidelity.

Quantum Computer Qubit Grid

The error correction algorithm must consider both accuracy and hardware structure. We chose IBM's Heron r1 processor (Fig. 1), which primarily uses CZ-gates, and has a median error rate of 0.004101.

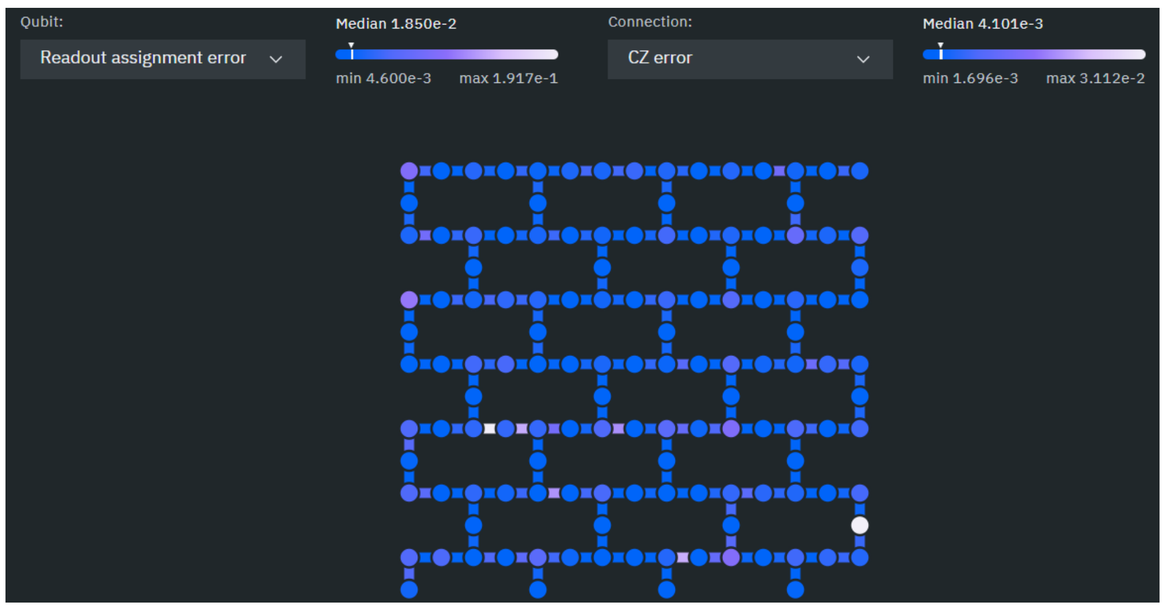

Fig. 1 The arragement of the qubits and the current error rate map of IBM Heron r1 processor

Grouping Error Correction in Circle

First, according to SRCC, the three-bit repetition code is to uniformly correct these three qubits to the majority answer, which means putting these three qubits into the full adder and taking the carry bit. In the experiment, we arranged those bits to be corrected into a ring shape and took out all three adjacent bits for three-bit repeated correction.

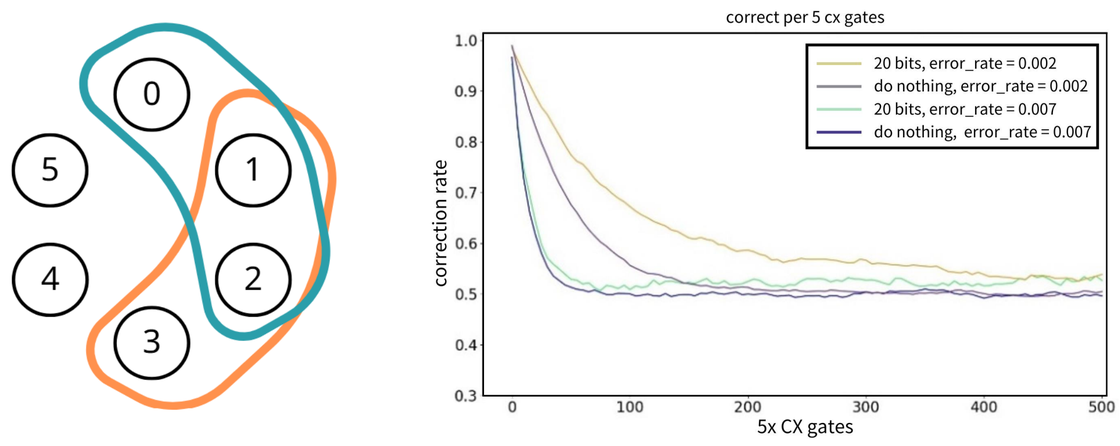

Fig. 2 Results of Grouping Error Correction in Circle. Consider the error rate equal to 0.007 and 0,002 in without error correction (dark blue and green) and SRCC erroron (purple and yellow).

Sampling According to Offset: [-2,0,1]

After experiments, we found that sampling according to offset: [-2,0,1] has the best error correction effect. The results are shown in Figure 3. We can see that the SRCC is useful in circle structure when the error rate below to 0.002 and in the best index.

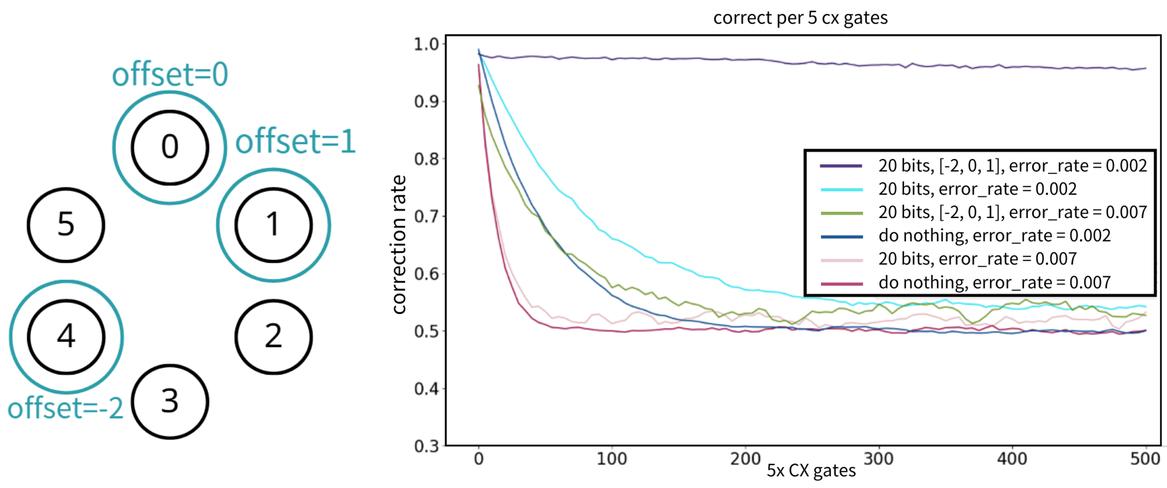

Fig. 3 Demonstration of how to take samples according to offset:[-2, 0, 1] and the results

CONCLUSION

Although none of the methods we proposed can maintain consistent accuracy at an error rate of 0.007, if we optimistically use the error rate of IBM's latest quantum computer (0.002) for experiments, our method can indeed maintain a certain accuracy. Our error correction method could maintain a certain level of accuracy on IBM's quantum computer. We also understand the importance of random permutation for all error correction algorithms.

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